Pseudo-activation-aware Quantization in DLMs

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Calibration Data in Quantization

Strengths:

- Determines the range,scale and zero-point for quantization
- Improves the output quality of quantized models
- Reduces reconstruction error

Weaknesses:

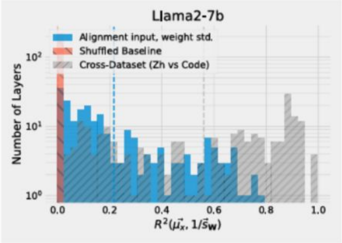
- The use of calibration data can lead to overfitting
- Can introduce bias if the calibration data is not representative

Pseudo-activation aware Quantization from Weight Structure

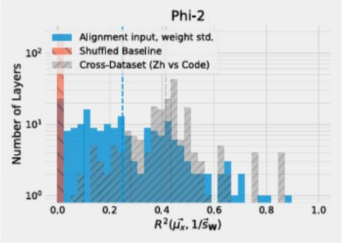
- SINQ is an uncalibrated quantization method, but at the same time it is an activation-aware quantization method.
- The mean of input activation and the weight standard deviation per column are strongly correlated
- Calibration data is not needed as we can directly use the weight standard deviation values to rescale the weight matrix before quantization

Pseudo-activation-aware Quantization from Weight Structure

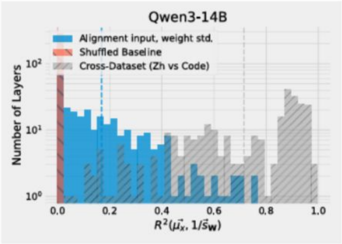
- This behavior holds for Qwen, Phi and Llama models.
- Does it hold for masked diffusion language models such as Llada and Dream?



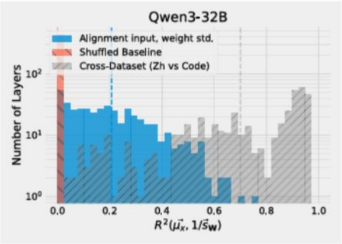
(a) Llama2-7b



(b) Phi-2

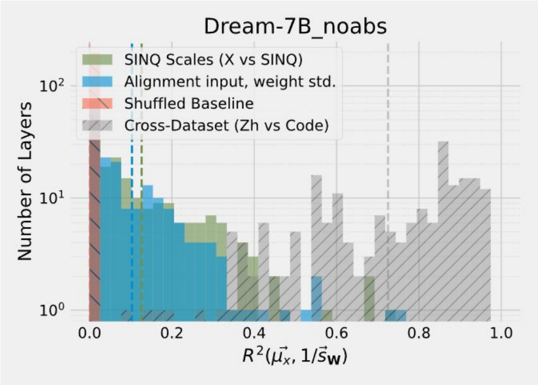


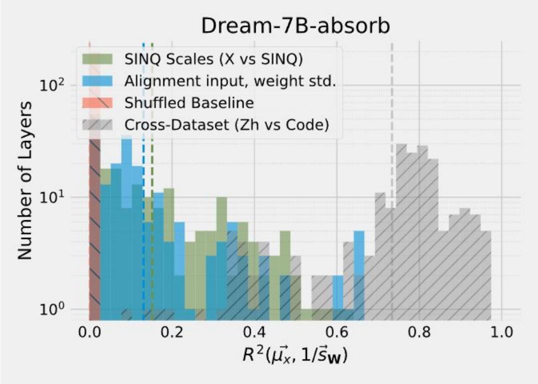
(c) Qwen3-14B

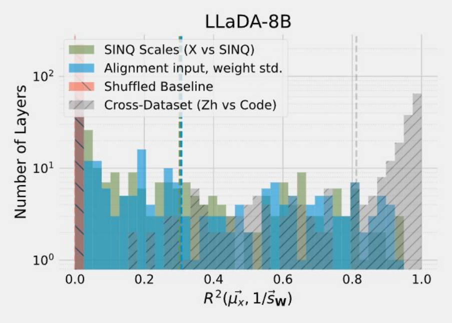


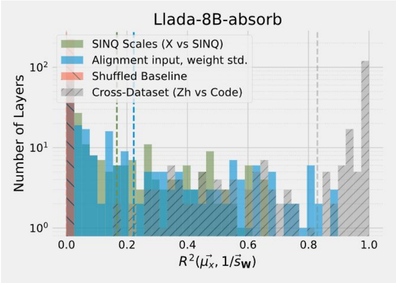
(d) Qwen3-32B

Pseudo-activation-aware Quantization from Weight Structure in DLMs









Masked Calibration Data

- Adding masking to calibration data may lead to more accurate activation statistics and correlation results

Algorithm 1 MCS: Masked Calibration Simulation

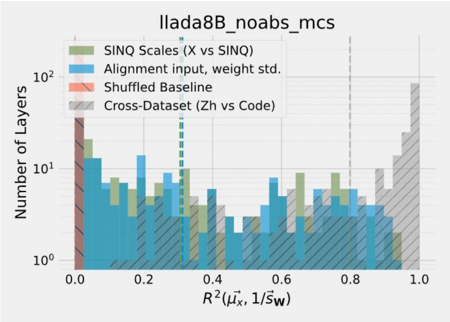
func MCS($\mathcal{X}, T, \gamma, \alpha(t)$)
Input: $\mathcal{X}$ — full sequences; T — total timesteps; γ — prefix ratio; $\alpha(t)$ — visibility schedule
Output: Masked calibration set $\mathcal{D}_{\text{calib}}$

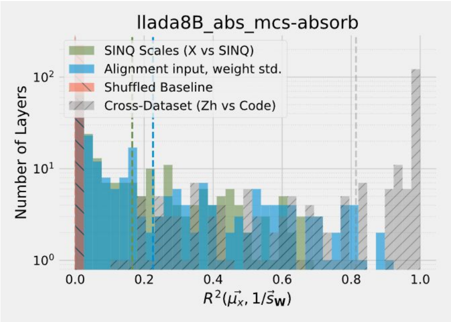
$\mathcal{D}_{\text{calib}} \leftarrow \emptyset$
for all $x \in \mathcal{X}$ **do**
    $L \leftarrow$ length of x
    $\mathcal{P} \leftarrow \{1, \ldots, \lfloor \gamma \cdot L \rfloor\}$
    for $t = 1$ to T **do**
        $\alpha \leftarrow \alpha(t)$
        for $i = 1$ to L **do**
            $r_i \leftarrow 1$ if $i \in \mathcal{P}$ else Bernoulli(α)
            $\tilde{x}_i(t) \leftarrow x_i$ if $r_i = 1$ else `[MASK]`
        end for
        $\mathcal{D}_{\text{calib}} \leftarrow \mathcal{D}_{\text{calib}} \cup \{\tilde{x}(t)\}$
    end for
end for
return $\mathcal{D}_{\text{calib}}$

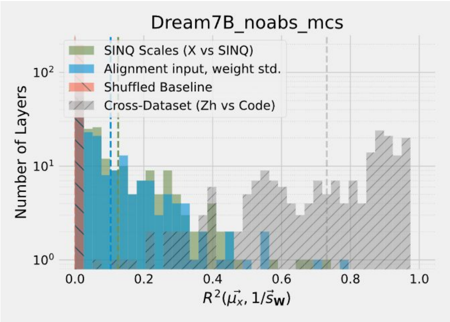
T. Zhang, Z. Li, X. Yan, H. Qin, Y. Guo, and Y. Zhang, “Quant-dLLM: Post-Training Extreme Low-Bit Quantization for Diffusion Large Language Models,” Sep. 27, 2025, *arXiv*: arXiv:2510.03274. doi: 10.48550/arXiv.2510.03274.

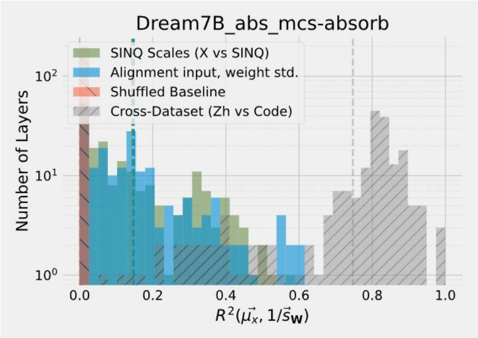
Masked Calibration Data

- Input activations are consistent across different datasets

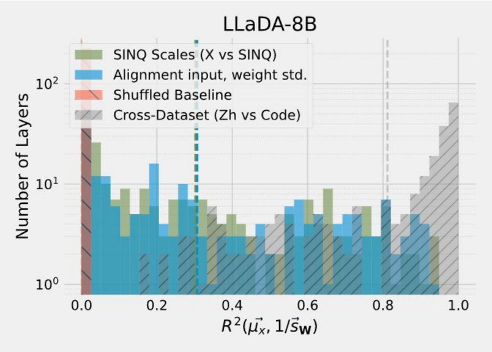


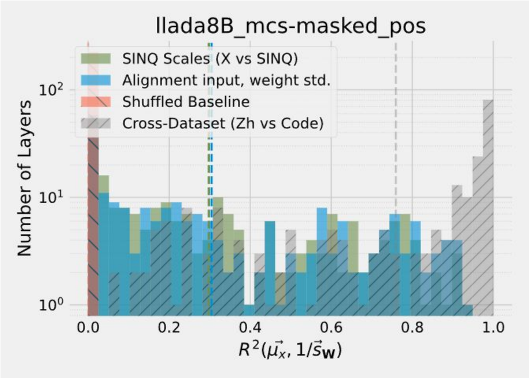


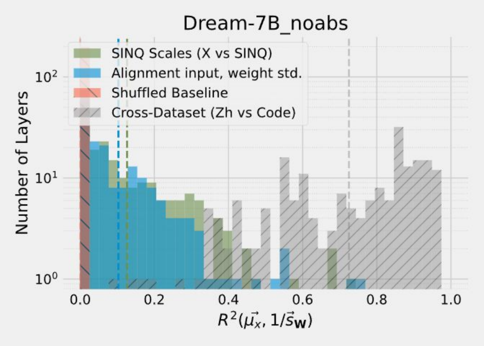


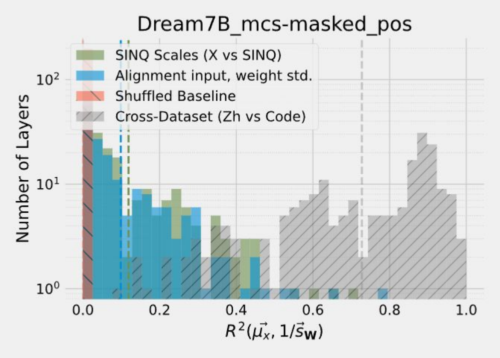


Correlation of Masked Positions Only



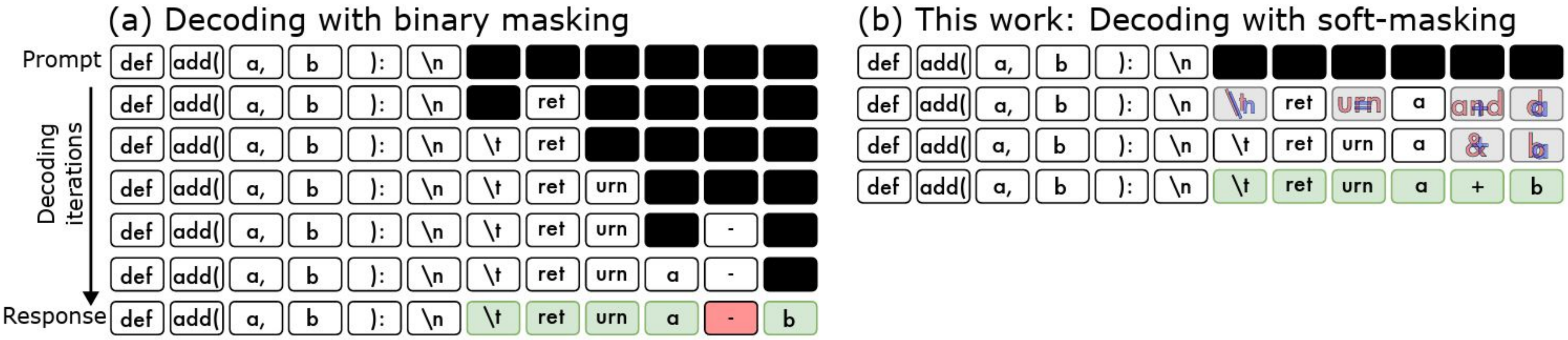






Soft Masked Diffusion Models

- Propagating the predictive information of masked tokens throughout the generation process



M. Hersche, S. Moor-Smith, T. Hofmann, and A. Rahimi, “Soft-Masked Diffusion Language Models,” Mar. 02, 2026, *arXiv*: arXiv:2510.17206. doi: [10.48550/arXiv.2510.17206](https://doi.org/10.48550/arXiv.2510.17206).

Plan for Next Week

- Analyze how quantization error accumulates across diffusion timesteps
- Aim to find a way to make SINQ aware of diffusion timesteps and prevent accumulation of error if it happens